Top 110 Interview Questions

AI, ML, DL, NLP, Data Science, ChatGPT, Claude
+ RAG, LangChain and LangGraph

Prepared for: Deloitte Cyber Security Role (AI and DS background)

Red text = Interview Question **Green text** = Simple Answer with 6 points and real-world examples

Contents

SECTION 1: ARTIFICIAL INTELLIGENCE (AI) (Q1 to Q15)

SECTION 2: MACHINE LEARNING (ML) (Q16 to Q40)

SECTION 3: DEEP LEARNING (DL) (Q41 to Q60)

SECTION 4: NATURAL LANGUAGE PROCESSING (NLP) (Q61 to Q72)

SECTION 5: DATA SCIENCE (DS) (Q73 to Q85)

SECTION 6: CHATGPT AND LARGE LANGUAGE MODELS (Q86 to Q93)

SECTION 7: CLAUDE (ANTHROPIC) (Q94 to Q100)

SECTION 8: RAG, LANGCHAIN AND LANGGRAPH (QUICK ROUND) (Q101 to Q110)

Tip: In each answer, say the one-line answer first, then add one or two real-world examples. Where you can, connect the topic to cyber security (for example, how it helps defence or how attackers misuse it). That shows the interviewer you can link your AI/DS knowledge to their field.

SECTION 1: ARTIFICIAL INTELLIGENCE (AI)

Q1. What is Artificial Intelligence?

Answer: Artificial Intelligence is the ability of a computer system to do tasks that normally need human intelligence, like learning, reasoning, and decision making.

- AI systems learn from data or rules instead of being told every step.
- Main goals of AI are learning, reasoning, perception, language understanding, and problem solving.
- AI can be rule-based (fixed logic) or learning-based (learns from data).
- Today most AI we use is Narrow AI, which is built for one specific task.
- **Real-world example:** Google Maps predicts traffic and suggests the fastest route using AI.
- **Real-world example:** Banks use AI to flag unusual card transactions as possible fraud in seconds.

Q2. What is the difference between AI, Machine Learning, and Deep Learning?

Answer: AI is the big idea of smart machines, Machine Learning is a part of AI that learns from data, and Deep Learning is a part of Machine Learning that uses many-layered neural networks.

- Think of three circles: AI is the biggest, ML sits inside AI, and DL sits inside ML.
- AI can work with simple if-else rules, but ML must have data to learn from.
- Classic ML needs humans to choose features, but DL finds features by itself.
- DL needs much more data and computing power (GPUs) than classic ML.
- **Real-world example:** A chess program with fixed rules is AI, a spam filter learning from emails is ML.
- **Real-world example:** Face unlock on your phone and ChatGPT are examples of Deep Learning.

Q3. What are the types of AI?

Answer: AI is grouped by capability into Narrow AI, General AI, and Super AI, and by function into reactive, limited memory, theory of mind, and self-aware AI.

- Narrow AI (Weak AI) does one task well, and it is the only type that exists today.
- General AI (AGI) would think and learn like a human across any task, but it does not exist yet.
- Super AI would be smarter than humans in every area, and it is only a theory.
- Reactive machines have no memory, and limited-memory AI uses past data to decide.
- **Real-world example:** Siri, Alexa, and Netflix recommendations are all Narrow AI.
- **Real-world example:** Self-driving cars use limited-memory AI because they remember recent road data.

Q4. What is the Turing Test?

Answer: The Turing Test, proposed by Alan Turing in 1950, checks whether a machine can talk so well that a human cannot tell if it is a machine.

- A human judge chats with a hidden human and a hidden machine.
- If the judge cannot reliably tell which is the machine, the machine passes.
- It measures behaviour, not real understanding or consciousness.
- Critics say a machine can pass just by imitating, without truly thinking.
- **Real-world example:** Modern chatbots like ChatGPT can often sound human, which brings the test back into discussion.
- **Real-world example:** CAPTCHA is called a reverse Turing test because it checks that you are a human, not a bot.

Q5. What is an intelligent agent in AI?

Answer: An intelligent agent is a program that senses its environment, makes decisions, and takes actions to reach a goal.

- It works in a loop: perceive, think, act, and then repeat.
- Agents have sensors (to observe) and actuators (to act).
- Types include simple reflex, model-based, goal-based, utility-based, and learning agents.
- Modern LLM agents can also use tools like search, code, and APIs.
- **Real-world example:** A robot vacuum senses walls and dirt and then decides where to clean.
- **Real-world example:** A security bot that watches logs and automatically blocks a suspicious IP is an agent.

Q6. What is the difference between rule-based systems and machine learning systems?

Answer: A rule-based system follows rules written by humans, while a machine learning system learns its own rules from data.

- Rule-based systems are easy to explain but hard to scale when rules grow.
- ML systems can handle new and complex patterns that humans did not write down.
- Rule-based systems cannot adapt, but ML models can be retrained with new data.
- ML needs good data, and its decisions can be harder to explain.
- **Real-world example:** A firewall rule saying "block port 23" is rule-based.
- **Real-world example:** An ML-based antivirus can catch a new malware it has never seen by spotting suspicious behaviour.

Q7. What is Generative AI?

Answer: Generative AI is AI that creates new content such as text, images, audio, code, or video by learning patterns from large amounts of data.

- It learns the pattern of data and then produces new data in a similar style.
- Common models are LLMs, GANs, diffusion models, and VAEs.
- It is often built on the Transformer architecture.
- It can make mistakes, invent facts, and be misused.
- **Real-world example:** ChatGPT and Claude write emails, code, and summaries.
- **Real-world example:** Attackers use generative AI to write convincing phishing emails and make deepfake voices.

Q8. What is bias in AI and how can we reduce it?

Answer: AI bias means the model gives unfair results for some groups because of unbalanced data or wrong design choices.

- Bias often comes from training data that does not represent everyone.
- It can also come from human labelling mistakes or from the model's design.
- Reduce it by using diverse data, checking fairness metrics, and testing on every group.
- Keep humans in the loop and audit models regularly.
- **Real-world example:** Amazon dropped an AI hiring tool that preferred male candidates because it learned from past male-heavy resumes.
- **Real-world example:** Some face recognition systems had higher error rates for darker skin, which led to wrong identifications.

Q9. What is Explainable AI (XAI)?

Answer: Explainable AI is a set of methods that help humans understand why an AI model made a particular decision.

- Many models, like deep neural networks, are "black boxes" that are hard to understand.
- Tools like SHAP and LIME show which input features pushed the decision.
- Explainability builds trust and helps meet laws and audits.
- It also helps security teams find if a model is being tricked.
- **Real-world example:** A bank must explain to a customer why a loan was rejected.
- **Real-world example:** A SOC analyst needs to know why the AI flagged a login as risky before blocking a user.

Q10. What is Responsible or Ethical AI?

Answer: Responsible AI means building and using AI in a way that is fair, safe, transparent, private, and accountable.

- Key principles are fairness, transparency, privacy, safety, and accountability.
- Companies create AI policies and review boards to follow these principles.
- Laws like the EU AI Act and data laws such as GDPR and India's DPDP Act push companies to follow them.
- Humans must remain responsible for AI decisions.
- **Real-world example:** Deloitte and other big firms have Trustworthy AI frameworks for their clients.
- **Real-world example:** Healthcare AI must protect patient data and be tested before real use.

Q11. What is AI hallucination?

Answer: AI hallucination is when an AI gives an answer that sounds confident but is false or made up.

- LLMs predict the next likely word and do not truly check facts.
- It happens more when the model lacks knowledge or the prompt is unclear.
- It can be reduced with RAG, better prompts, and human review.
- Always verify important facts from trusted sources.
- **Real-world example:** A lawyer in the US was penalised after ChatGPT invented fake court cases in his filing.
- **Real-world example:** A chatbot may suggest a software package that does not exist, and attackers can register that name with malware inside.

Q12. How is AI used in cybersecurity?

Answer: AI helps security teams detect threats faster, reduce false alarms, and respond automatically, because it can analyse huge amounts of data quickly.

- Anomaly detection finds unusual behaviour in logs and networks.
- Malware and phishing detection uses ML to spot bad files and emails.
- UEBA tracks user behaviour to catch insider threats and stolen accounts.
- SOAR and AI assistants automate response and summarise alerts.
- **Real-world example:** Microsoft Defender and CrowdStrike use ML to detect malware by behaviour.
- **Real-world example:** Gmail's AI blocks millions of phishing and spam emails every day.

Q13. What are adversarial attacks on AI?

Answer: Adversarial attacks are tricks where an attacker slightly changes the input to make an AI model give a wrong answer.

- The change is often so small that humans cannot notice it.
- Evasion attacks fool the model at the time of use.
- Poisoning attacks corrupt the training data.
- Defences include adversarial training, input checking, and monitoring.
- **Real-world example:** Stickers placed on a stop sign made an AI vision system read it as a speed limit sign.
- **Real-world example:** Attackers change malware code slightly so an ML antivirus marks it as safe.

Q14. What are the main risks of using AI in an enterprise?

Answer: The main risks are data leakage, wrong or biased outputs, model attacks, legal non-compliance, and over-dependence on AI.

- Employees may paste confidential data into public AI tools.
- Models can be attacked through prompt injection, poisoning, or theft.
- Wrong outputs can cause bad business decisions.
- Rules like GDPR and the EU AI Act can bring heavy fines if ignored.
- **Real-world example:** Samsung employees leaked source code by pasting it into ChatGPT in 2023.
- **Real-world example:** Many companies now publish an AI usage policy and block unapproved AI tools.

Q15. What is Agentic AI?

Answer: Agentic AI is AI that can plan, take multi-step actions, and use tools on its own to complete a goal with little human help.

- Normal chatbots only answer, but agents can also act, like search, run code, or send emails.
- Agents break a goal into steps, run the steps, check results, and repeat.
- They bring new risks because a wrong action can have real impact.
- Good controls are limited permissions, logging, and human approval for risky steps.
- **Real-world example:** A coding agent like Claude Code can read a project, edit files, and run tests.
- **Real-world example:** A security agent can investigate an alert, gather logs, and write a ticket automatically.

SECTION 2: MACHINE LEARNING (ML)

Q16. What is Machine Learning?

Answer: Machine Learning is a part of AI where computers learn patterns from data and improve without being clearly programmed for every case.

- The model is trained on data, and then it makes predictions on new data.
- Basic steps are collect data, clean it, train, test, and deploy.
- Performance improves as it sees more good-quality data.
- Main types are supervised, unsupervised, and reinforcement learning.
- **Real-world example:** Netflix recommends shows by learning from what you and others watch.
- **Real-world example:** Email services learn to detect spam from millions of marked emails.

Q17. What are supervised, unsupervised, and reinforcement learning?

Answer: Supervised learning uses labelled data, unsupervised learning finds patterns in unlabelled data, and reinforcement learning learns by trial and error with rewards.

- Supervised has an answer for each example, such as spam or not spam.
- Unsupervised has no answers, so the model groups or compresses the data.
- Reinforcement learning uses an agent that gets rewards or penalties for its actions.
- Supervised does classification and regression, and unsupervised does clustering.
- **Real-world example:** Predicting house prices is supervised learning.
- **Real-world example:** Grouping customers by buying habits is unsupervised, and a robot learning to walk is reinforcement learning.

Q18. What is Reinforcement Learning?

Answer: Reinforcement Learning is a method where an agent learns the best actions by trying things and getting rewards for good results and penalties for bad ones.

- Main parts are agent, environment, action, state, and reward.
- The goal is to get the highest total reward over time.
- It balances exploration (trying new things) and exploitation (using what works).
- It needs lots of trials, so it is often trained in simulation.
- **Real-world example:** DeepMind's AlphaGo beat the world Go champion using reinforcement learning.
- **Real-world example:** ChatGPT uses RLHF, which is reinforcement learning from human feedback, to give better answers.

Q19. What are overfitting and underfitting?

Answer: Overfitting is when a model memorises the training data and fails on new data, and underfitting is when a model is too simple to learn the pattern at all.

- Overfitting shows high training accuracy but low test accuracy.
- Underfitting shows low accuracy on both training and test data.
- Fix overfitting with more data, regularisation, dropout, or a simpler model.
- Fix underfitting with a more complex model or better features.
- **Real-world example:** A student who memorises answers without understanding fails a new question, which is overfitting.
- **Real-world example:** A fraud model trained on only one month of data may overfit and miss new fraud patterns.

Q20. What is the bias-variance tradeoff?

Answer: Bias is error from a model being too simple, variance is error from a model being too sensitive to training data, and we must balance both.

- High bias leads to underfitting, and high variance leads to overfitting.
- Making a model more complex lowers bias but raises variance.
- The goal is a model with low total error on new data.
- Cross-validation and regularisation help find the balance.
- **Real-world example:** A straight line for predicting stock prices is high bias because it is too simple.
- **Real-world example:** A very deep decision tree that fits every data point is high variance.

Q21. What is the difference between training, validation, and test data?

Answer: Training data teaches the model, validation data tunes the model, and test data gives a final honest score.

- A common split is 70 percent training, 15 percent validation, and 15 percent test.
- The test set must never be used while building the model.
- The validation set helps choose hyperparameters and compare models.
- Mixing these sets gives fake high accuracy.
- **Real-world example:** Practice lessons are training, a mock test is validation, and the final exam is the test set.
- **Real-world example:** Kaggle keeps a hidden test set so nobody can cheat on the leaderboard.

Q22. What is cross-validation?

Answer: Cross-validation is a method that splits data into several parts, trains on some and tests on the rest, and repeats this to get a reliable score.

- In k-fold, data is split into k parts, and each part is used once as the test set.
- The final score is the average of all k results.
- It reduces the risk of a lucky or unlucky single split.
- It is very useful when the dataset is small.
- **Real-world example:** A hospital with only 500 patient records uses 5-fold cross-validation to test a disease model.
- **Real-world example:** Data scientists use it to choose between models like Random Forest and SVM fairly.

Q23. What is regularisation (L1 and L2)?

Answer: Regularisation adds a penalty to the model for being too complex, which helps stop overfitting.

- L1 (Lasso) can shrink some weights to exactly zero, so it also selects features.
- L2 (Ridge) shrinks all weights but rarely makes them zero.
- A hyperparameter (lambda) controls how strong the penalty is.
- Dropout is another form of regularisation used in neural networks.
- **Real-world example:** Credit scoring models use L1 to keep only the most useful customer features.
- **Real-world example:** Ridge regression is used when many inputs are related to each other, like in house price data.

Q24. What is Linear Regression?

Answer: Linear Regression predicts a number by fitting a straight line through the data points.

- The formula is y = mx + c, where m is the slope and c is the intercept.
- It finds the line that makes the total squared error smallest.
- It assumes a straight-line relationship between inputs and output.
- It is simple, fast, and easy to explain.
- **Real-world example:** Predicting house price from the area of the house.
- **Real-world example:** Estimating next month's sales from advertising spend.

Q25. What is Logistic Regression?

Answer: Logistic Regression is a classification method that predicts the probability of a class, such as yes or no, using the sigmoid function.

- Despite the name, it is used for classification and not for predicting numbers.
- The sigmoid function squeezes any value into a range between 0 and 1.
- A threshold, often 0.5, decides the final class.
- It is simple and gives understandable results.
- **Real-world example:** Predicting whether an email is spam or not.
- **Real-world example:** Predicting whether a login attempt is a fraud or a genuine user.

Q26. What is a Decision Tree?

Answer: A Decision Tree is a model that makes decisions by asking a series of yes or no questions, like a flowchart.

- Each node asks a question about a feature, and each leaf gives the answer.
- It picks questions using measures like Gini impurity or entropy.
- It is easy to read and explain.
- A single deep tree often overfits.
- **Real-world example:** A bank uses a tree to decide loan approval based on income, age, and credit score.
- **Real-world example:** A tree can classify network traffic as normal or an attack based on packet features.

Q27. What is a Random Forest?

Answer: A Random Forest is a group of many decision trees whose votes are combined to give a stronger and more stable answer.

- Each tree is trained on a random sample of data and random features.
- For classification it takes the majority vote, and for regression the average.
- It reduces overfitting compared with a single tree.
- It also shows which features are most important.
- **Real-world example:** Used in credit card fraud detection for its good accuracy.
- **Real-world example:** Used in intrusion detection systems, such as on the NSL-KDD dataset.

Q28. What is a Support Vector Machine (SVM)?

Answer: SVM is a classifier that finds the best boundary line that separates classes with the widest possible gap.

- The boundary is called a hyperplane, and the closest points are support vectors.
- A wider margin usually means better results on new data.
- The kernel trick lets SVM handle data that is not separable by a straight line.
- It works well on small or medium datasets with many features.
- **Real-world example:** SVM is used for classifying text such as news topics and spam.
- **Real-world example:** It is used in malware classification and in cancer detection from gene data.

Q29. What is K-Nearest Neighbours (KNN)?

Answer: KNN classifies a new point by looking at its K closest points and choosing the most common class among them.

- It does not build a model, so it is called a lazy learner.
- Distance measures like Euclidean distance find the closest points.
- Choosing K is important: too small gives noise, too large gives blurry results.
- Data should be scaled, and it gets slow with large data.
- **Real-world example:** Recommending movies based on users with similar tastes.
- **Real-world example:** Detecting unusual network connections that sit far from normal neighbours.

Q30. What is K-Means clustering?

Answer: K-Means groups data into K clusters so that points in the same cluster are close to each other.

- You choose K, and the algorithm places K centre points randomly.
- It assigns each point to the nearest centre and then moves the centres to the average, repeating until stable.
- It is unsupervised, so no labels are needed.
- The elbow method helps choose a good K.
- **Real-world example:** Retailers group customers into segments such as budget and premium buyers.
- **Real-world example:** Security teams cluster similar malware samples into families.

Q31. What is Naive Bayes?

Answer: Naive Bayes is a fast classifier based on Bayes' theorem that assumes every feature is independent of the others.

- It calculates the probability of each class given the input words or features.
- The word "naive" is used because the independence assumption is rarely fully true.
- It works very well for text with little training data.
- It is fast and simple to build.
- **Real-world example:** Early spam filters like SpamAssassin used Naive Bayes.
- **Real-world example:** It is used for quick sentiment analysis of product reviews.

Q32. What is Gradient Boosting and XGBoost?

Answer: Gradient Boosting builds trees one after another, where each new tree fixes the mistakes of the earlier ones, and XGBoost is a fast and powerful version of it.

- It combines many weak models into one strong model.
- XGBoost adds regularisation, parallel work, and handling of missing values.
- It often wins competitions on tabular data.
- It needs careful tuning to avoid overfitting.
- **Real-world example:** Many Kaggle winners use XGBoost or LightGBM on structured data.
- **Real-world example:** Banks use boosted models for credit risk and fraud scoring.

Q33. What is the difference between Bagging and Boosting?

Answer: Bagging trains many models in parallel on random samples and averages them, while boosting trains models one by one where each fixes the previous model's errors.

- Bagging mainly reduces variance (overfitting).
- Boosting mainly reduces bias (underfitting).
- Random Forest is a bagging method, and AdaBoost and XGBoost are boosting methods.
- Boosting can overfit on noisy data more easily.
- **Real-world example:** Random Forest (bagging) is used for stable fraud detection.
- **Real-world example:** XGBoost (boosting) is used for ranking and click-through prediction in ads.

Q34. What is PCA (Principal Component Analysis)?

Answer: PCA is a method that reduces the number of features while keeping most of the useful information in the data.

- It creates new features called principal components that capture the most variation.
- It removes noise and makes training faster.
- It helps to visualise high-dimensional data in 2D or 3D.
- The new features are harder to explain than the original ones.
- **Real-world example:** Compressing image data before feeding it to a model.
- **Real-world example:** Reducing hundreds of network traffic features before anomaly detection.

Q35. What are the confusion matrix, precision, recall, and F1-score?

Answer: A confusion matrix counts correct and wrong predictions, precision tells how many predicted positives were right, and recall tells how many real positives were found.

- True Positive, False Positive, True Negative, and False Negative are its four parts.
- Precision = TP / (TP + FP), and recall = TP / (TP + FN).
- F1-score is the balance (harmonic mean) of precision and recall.
- Accuracy alone is misleading when classes are imbalanced.
- **Real-world example:** In cancer detection, high recall matters because missing a patient is dangerous.
- **Real-world example:** In a SOC, low precision causes too many false alerts and analyst fatigue.

Q36. What is the ROC curve and AUC?

Answer: The ROC curve shows how well a classifier separates classes at different thresholds, and AUC is the area under it, where a higher value means a better model.

- It plots True Positive Rate against False Positive Rate.
- AUC of 0.5 means random guessing, and 1.0 means a perfect model.
- It helps compare models without choosing a single threshold.
- For very imbalanced data, a Precision-Recall curve is often better.
- **Real-world example:** Comparing two phishing detectors by their AUC scores.
- **Real-world example:** Medical test models are judged on AUC before use in hospitals.

Q37. How do you handle imbalanced data?

Answer: Imbalanced data means one class is much bigger than the other, and we handle it by balancing the data, using better metrics, or adjusting the model.

- Oversample the small class, for example with SMOTE, or undersample the big class.
- Use class weights so the model pays more attention to the rare class.
- Judge using precision, recall, F1, or AUC instead of accuracy.
- Collect more data of the rare class if possible.
- **Real-world example:** Fraud is under 1 percent of transactions, so a model that says "no fraud" always is 99 percent accurate but useless.
- **Real-world example:** Attack traffic is rare in a network, so intrusion models need special balancing.

Q38. What is feature engineering and feature selection?

Answer: Feature engineering is creating useful inputs from raw data, and feature selection is choosing only the most useful inputs for the model.

- Good features often matter more than the choice of algorithm.
- Examples are extracting day or hour from a date, or making ratios.
- Feature selection removes useless features, which reduces overfitting and training time.
- Methods include correlation, L1 regularisation, and feature importance.
- **Real-world example:** For fraud detection, creating a feature like "number of transactions in the last hour".
- **Real-world example:** For phishing detection, features like URL length and presence of "@" in the link.

Q39. What is data leakage?

Answer: Data leakage is when information from outside the training data, often from the test set or the future, sneaks into training and makes results look better than they really are.

- It gives very high test scores but poor real-world performance.
- Common cause is scaling or cleaning the full dataset before splitting it.
- Another cause is including a feature that reveals the answer.
- Prevent it by splitting first and using pipelines.
- **Real-world example:** A hospital model that used "treatment given" as a feature to predict disease looked perfect but failed in practice.
- **Real-world example:** Using future stock prices while training a prediction model.

Q40. What is data poisoning in machine learning?

Answer: Data poisoning is an attack where someone adds bad or fake data into the training set so that the model learns wrong behaviour.

- It targets the training stage, not the live use of the model.
- A backdoor attack hides a secret trigger that makes the model give a wrong answer later.
- Defences are checking data sources, anomaly checks on data, and access control.
- Keep a record of where data came from (data provenance).
- **Real-world example:** Microsoft's Tay chatbot (2016) was taught offensive language by users within hours.
- **Real-world example:** Attackers can upload many fake samples to a public dataset so that a malware detector learns to trust their malware.

SECTION 3: DEEP LEARNING (DL)

Q41. What is Deep Learning?

Answer: Deep Learning is a type of machine learning that uses neural networks with many layers to learn complex patterns from large amounts of data.

- The word "deep" means many hidden layers in the network.
- It learns features automatically, so little manual feature work is needed.
- It needs large data and strong hardware like GPUs.
- It is the base of modern vision, speech, and language AI.
- **Real-world example:** Face unlock on smartphones uses deep learning.
- **Real-world example:** Google Translate and voice assistants use deep neural networks.

Q42. What is a neural network and how does it work?

Answer: A neural network is a group of connected nodes (neurons) arranged in layers that learns by adjusting the strength of connections, called weights, to reduce its errors.

- It has an input layer, one or more hidden layers, and an output layer.
- Each neuron multiplies inputs by weights, adds a bias, and applies an activation function.
- The network makes a prediction, measures the error, and then updates weights.
- This repeats over many rounds (epochs) until the error is small.
- **Real-world example:** A network learns to recognise handwritten digits from thousands of examples.
- **Real-world example:** Neural networks detect abnormal behaviour in user login patterns.

Q43. What are activation functions?

Answer: An activation function decides whether a neuron should fire and adds non-linearity so the network can learn complex patterns.

- Without it, the whole network would act like one simple linear model.
- ReLU gives zero for negative values and the same value for positive ones, and it is the most common in hidden layers.
- Sigmoid gives values from 0 to 1 and is used for yes or no outputs.
- Softmax turns outputs into probabilities for many classes, and tanh gives values from -1 to 1.
- **Real-world example:** Image classifiers use ReLU inside and Softmax at the end to pick one of many classes.
- **Real-world example:** A spam detector uses Sigmoid at the output to give a spam probability.

Q44. What is backpropagation?

Answer: Backpropagation is the method that sends the error backward through the network to find how much each weight caused the error, so the weights can be corrected.

- First, forward pass calculates the prediction and the loss.
- Then, backward pass uses the chain rule to calculate gradients for each weight.
- An optimizer uses these gradients to update the weights.
- It is the main reason deep networks can be trained.
- **Real-world example:** Every time ChatGPT-like models are trained, backpropagation updates billions of weights.
- **Real-world example:** Training an image model to recognise cats, the error from wrong guesses is pushed back to improve it.

Q45. What is gradient descent and what are its types?

Answer: Gradient Descent is an optimisation method that moves the weights step by step in the direction that reduces the loss, like walking downhill.

- The learning rate controls the size of each step.
- Batch gradient descent uses all the data for each update, and it is slow.
- Stochastic gradient descent (SGD) uses one sample, and it is noisy but fast.
- Mini-batch uses small groups of data, and it is the most common choice.
- **Real-world example:** Imagine walking down a foggy mountain by always stepping in the steepest downward direction.
- **Real-world example:** Almost all deep learning frameworks train models using mini-batch gradient descent.

Q46. What are the vanishing and exploding gradient problems?

Answer: Vanishing gradients make the updates so small that early layers stop learning, and exploding gradients make updates so big that training becomes unstable.

- They happen in very deep networks or long sequences because gradients are multiplied many times.
- Sigmoid and tanh functions often cause the vanishing problem.
- Fixes include ReLU, batch normalisation, gradient clipping, and residual connections.
- LSTM and Transformers were designed partly to solve this.
- **Real-world example:** Early RNNs could not remember words from the start of a long sentence because of vanishing gradients.
- **Real-world example:** ResNet used skip connections to train networks with over 100 layers.

Q47. What is a loss function? Name some common ones.

Answer: A loss function measures how wrong the model's predictions are, and training tries to make this number as small as possible.

- Mean Squared Error (MSE) is used for predicting numbers.
- Binary cross-entropy is used for two-class problems.
- Categorical cross-entropy is used for many-class problems.
- Choosing the right loss is important for the right behaviour.
- **Real-world example:** House price prediction uses MSE.
- **Real-world example:** Classifying emails as spam or not uses binary cross-entropy.

Q48. What is a Convolutional Neural Network (CNN)?

Answer: A CNN is a neural network designed for images that uses small filters to detect patterns like edges, shapes, and objects.

- Convolution layers slide filters over the image to find features.
- Pooling layers shrink the data and keep important information.
- Early layers detect simple edges, and later layers detect complex objects.
- It shares weights, so it needs fewer parameters than a normal network.
- **Real-world example:** Hospitals use CNNs to detect tumours in X-rays and MRI scans.
- **Real-world example:** Malware researchers convert malware files into images and classify them with CNNs.

Q49. What is a Recurrent Neural Network (RNN)?

Answer: An RNN is a neural network for sequence data that keeps a memory of earlier inputs while reading new ones.

- It processes data step by step, such as word by word.
- The hidden state carries information from the past.
- It struggles with long sequences because of vanishing gradients.
- Transformers have mostly replaced RNNs for language tasks.
- **Real-world example:** Early speech recognition and text prediction used RNNs.
- **Real-world example:** Predicting the next value in a time series, such as network traffic over time.

Q50. What is LSTM?

Answer: LSTM (Long Short-Term Memory) is a special RNN with gates that decide what to remember and what to forget, so it can learn long-term patterns.

- It has three gates: input, forget, and output.
- A memory cell carries information over long sequences.
- It reduces the vanishing gradient problem.
- GRU is a simpler and faster version of LSTM.
- **Real-world example:** Used in stock price and weather forecasting.
- **Real-world example:** Used to detect attacks by learning normal patterns in system log sequences.

Q51. What is the Transformer architecture and the attention mechanism?

Answer: A Transformer is a neural network that uses self-attention to look at all words in a sentence at once and decide which words are most important to each other.

- It was introduced in the 2017 paper "Attention Is All You Need."
- Self-attention lets a word connect to other words, no matter how far apart.
- It processes words in parallel, so training is much faster than RNNs.
- It is the base of GPT, BERT, Claude, and most modern LLMs.
- **Real-world example:** In "The bank by the river," attention helps the model know that "bank" means a river bank.
- **Real-world example:** ChatGPT, Claude, and Google Gemini are all built on Transformers.

Q52. What are dropout and batch normalisation?

Answer: Dropout randomly turns off some neurons during training to prevent overfitting, and batch normalisation scales layer outputs to make training faster and more stable.

- Dropout forces the network not to depend on any single neuron.
- Dropout is used only during training and not during prediction.
- Batch normalisation keeps values in a steady range between layers.
- Both help the model to generalise better.
- **Real-world example:** Adding 50 percent dropout in a image classifier reduces overfitting.
- **Real-world example:** Batch normalisation lets very deep networks like ResNet train faster.

Q53. What is an optimizer? Explain Adam.

Answer: An optimizer is the algorithm that updates the model's weights to reduce loss, and Adam is a popular one that adapts the learning rate for each weight.

- SGD is simple but can be slow and may get stuck.
- Adam combines momentum and adaptive learning rates.
- It works well with little tuning, so it is the default for many projects.
- Others include RMSprop and AdaGrad.
- **Real-world example:** Adam is used to train most Transformer and language models.
- **Real-world example:** Beginners use Adam in Keras or PyTorch for quick and stable results.

Q54. What is transfer learning?

Answer: Transfer learning means reusing a model already trained on a big dataset and adjusting it for your own smaller task.

- It saves time, data, and computing cost.
- You keep the early layers and retrain the last layers for your task.
- Fine-tuning is a common form of transfer learning.
- Pretrained models such as ResNet, BERT, and GPT are widely reused.
- **Real-world example:** A hospital fine-tunes ImageNet-trained ResNet to detect pneumonia with only a few thousand X-rays.
- **Real-world example:** A company fine-tunes BERT to classify its own support tickets.

Q55. What is a GAN (Generative Adversarial Network)?

Answer: A GAN is made of two networks, a generator that creates fake data and a discriminator that tries to catch fakes, and both improve by competing.

- The generator tries to fool the discriminator.
- The discriminator learns to tell real from fake.
- Over time the fake data becomes very realistic.
- GANs are hard to train and can be unstable.
- **Real-world example:** Used to create realistic fake faces on sites like "This Person Does Not Exist."
- **Real-world example:** Used to make deepfake videos, which is a major social engineering and fraud risk.

Q56. What is an autoencoder?

Answer: An autoencoder is a neural network that compresses data into a small form and then rebuilds it, and it learns the important features while doing so.

- It has an encoder (compress) and a decoder (rebuild).
- It is trained to make the output close to the input.
- Used for noise removal, compression, and anomaly detection.
- Big rebuild error on new data means that data is unusual.
- **Real-world example:** Autoencoders detect unusual network traffic because attack data is rebuilt poorly.
- **Real-world example:** Used to clean noisy photos and scanned documents.

Q57. What are epoch, batch size, iteration, and hyperparameters?

Answer: An epoch is one full pass over the training data, batch size is how many samples are processed together, and hyperparameters are settings we choose before training.

- Iteration is one update step, so iterations per epoch equals data size divided by batch size.
- Learning rate, number of layers, and dropout rate are hyperparameters.
- Weights are learned by the model, but hyperparameters are set by us.
- Tuning can be done by grid search, random search, or Bayesian methods.
- **Real-world example:** 1000 images with batch size 100 means 10 iterations per epoch.
- **Real-world example:** Teams tune the learning rate to speed up training without losing accuracy.

Q58. What are adversarial examples in deep learning and how do we defend against them?

Answer: Adversarial examples are inputs with tiny hidden changes that cause a deep learning model to give a wrong result, while humans see no difference.

- They exploit the way neural networks respond to small pixel or feature changes.
- Attacks like FGSM and PGD create them using the model's gradients.
- Defences include adversarial training, input filtering, and model ensembles.
- Testing models with "red teaming" before release is a good practice.
- **Real-world example:** A small sticker pattern made a vision model misread road signs.
- **Real-world example:** Slightly edited malware bytes can pass an ML malware scanner.

Q59. What are deepfakes and how can they be detected?

Answer: Deepfakes are fake videos, images, or audio made by deep learning that look or sound like a real person.

- They are made mostly with GANs, autoencoders, and diffusion models.
- Detection looks for odd blinking, lighting errors, mouth mismatch, and digital traces.
- Detection models and watermarking (such as C2PA content credentials) help.
- Staff training and call-back verification are strong defences.
- **Real-world example:** In 2024, a Hong Kong employee of a firm was tricked into transferring about 25 million dollars after a deepfake video call.
- **Real-world example:** Voice cloning is used in "CEO fraud" calls asking for urgent payments.

Q60. What is the difference between TensorFlow and PyTorch?

Answer: Both are deep learning frameworks, where TensorFlow (by Google) is strong for production deployment and PyTorch (by Meta) is popular for research and easy debugging.

- PyTorch uses dynamic graphs, so code feels like normal Python.
- TensorFlow offers tools like TensorFlow Serving, Lite, and TFX for deployment.
- Keras is a simple high-level API that runs on TensorFlow.
- Both support GPUs and are free and open source.
- **Real-world example:** Many research papers and Hugging Face models use PyTorch.
- **Real-world example:** Google products and mobile apps often use TensorFlow Lite.

SECTION 4: NATURAL LANGUAGE PROCESSING (NLP)

Q61. What is Natural Language Processing?

Answer: NLP is the part of AI that helps computers understand, interpret, and generate human language.

- It covers text and speech tasks.
- Common tasks are translation, sentiment analysis, summarising, and question answering.
- It combines linguistics, machine learning, and deep learning.
- Modern NLP is dominated by Transformer-based models.
- **Real-world example:** Google Translate converts text between over 100 languages.
- **Real-world example:** Chatbots on bank websites understand customer questions using NLP.

Q62. What are tokenization, stemming, and lemmatization?

Answer: Tokenization splits text into small pieces called tokens, while stemming and lemmatization reduce words to their base forms.

- Tokens can be words, sub-words, or characters.
- Stemming cuts word endings by rules, like "running" to "run," and it can be rough.
- Lemmatization uses a dictionary to give a real base word, like "better" to "good."
- These steps clean text before it is given to a model.
- **Real-world example:** A search engine treats "running" and "runs" the same by reducing them to "run."
- **Real-world example:** ChatGPT breaks your sentence into sub-word tokens before processing.

Q63. What are stop words, Bag of Words, and TF-IDF?

Answer: Stop words are very common words like "the" and "is" that are often removed, Bag of Words counts the words in a text, and TF-IDF gives higher weight to words that are important and rare.

- Bag of Words ignores the word order and only counts words.
- TF means how often a word is in a document, and IDF lowers the weight of words common in all documents.
- A high TF-IDF word is special to that document.
- These are simple and fast methods for text classification.
- **Real-world example:** Spam filters use word counts to spot spam words like "lottery" and "winner."
- **Real-world example:** Search engines use TF-IDF style scoring to rank relevant pages.

Q64. What are word embeddings (Word2Vec)?

Answer: Word embeddings turn words into lists of numbers (vectors) so that words with similar meanings have similar numbers.

- Computers cannot understand words, so we convert them into vectors.
- Word2Vec and GloVe are classic methods.
- The famous example is king minus man plus woman gives a vector near queen.
- Modern models use contextual embeddings, which change with the sentence.
- **Real-world example:** Embeddings power semantic search, where "car" also finds "automobile."
- **Real-world example:** RAG systems store document embeddings in a vector database.

Q65. What are Named Entity Recognition (NER) and POS tagging?

Answer: NER finds names of people, places, and organisations in text, and POS tagging labels each word as a noun, verb, adjective, and so on.

- NER pulls out useful facts from unstructured text.
- POS tagging helps understand the grammar of a sentence.
- Libraries like spaCy and NLTK offer both.
- They are early steps in many NLP pipelines.
- **Real-world example:** News apps use NER to tag companies and places in articles.
- **Real-world example:** Security tools use NER to pick IP addresses, domains, and malware names from threat reports.

Q66. What is sentiment analysis?

Answer: Sentiment analysis finds whether a piece of text expresses a positive, negative, or neutral opinion.

- It can be rule-based, machine learning-based, or use pre-trained models like BERT.
- Sarcasm and mixed feelings make it difficult.
- Results can be a label or a score.
- Companies use it to track brand health at scale.
- **Real-world example:** Amazon and Flipkart analyse product reviews to find common complaints.
- **Real-world example:** Brands track tweets to catch a PR problem early.

Q67. What are BERT and GPT, and how are they different?

Answer: BERT reads text in both directions to understand it, while GPT reads left to right to generate text.

- BERT uses only the Transformer encoder and is great for understanding tasks like classification.
- GPT uses only the decoder and is great for generating text.
- BERT is trained by guessing hidden words, and GPT is trained by predicting the next word.
- Both are pretrained on huge text and then fine-tuned or prompted.
- **Real-world example:** Google Search uses BERT to understand search queries better.
- **Real-world example:** ChatGPT is based on the GPT family and writes essays and code.

Q68. What is a Large Language Model (LLM)?

Answer: An LLM is a very large neural network, trained on huge amounts of text, that can understand and generate human-like language.

- It has billions of parameters and is mostly based on Transformers.
- It learns by predicting the next token in text.
- It can do many tasks, such as summarising, translating, coding, and answering questions, without separate training for each.
- It can hallucinate and may reflect bias in its training data.
- **Real-world example:** GPT-4, Claude, Gemini, and Llama are all LLMs.
- **Real-world example:** A SOC uses an LLM to summarise a long incident report in seconds.

Q69. What are tokens and a context window?

Answer: A token is a small piece of text that an LLM reads, and the context window is the maximum number of tokens the model can handle at one time.

- One token is roughly three-fourths of an English word.
- The context window includes both your input and the model's output.
- If the text is too long, older parts are cut or forgotten.
- Pricing of LLM APIs is usually based on tokens.
- **Real-world example:** Uploading a long PDF to Claude works because it has a large context window.
- **Real-world example:** A very long chat with an LLM may start to forget the early instructions.

Q70. How is NLP used to detect phishing and spam?

Answer: NLP reads the email text, subject, and links to spot signs of fraud, such as urgent language, fake sender style, and suspicious words.

- Text is converted to features using TF-IDF or embeddings.
- A classifier like Naive Bayes, SVM, or BERT gives a phishing score.
- It also checks the URL patterns and the writing tone.
- New LLM-written phishing needs stronger models and extra signals like sender reputation.
- **Real-world example:** Gmail and Microsoft 365 filter phishing mails using NLP and ML.
- **Real-world example:** Security awareness tools use NLP to check if a reported email is likely phishing.

Q71. What are the main challenges in NLP?

Answer: Language is full of ambiguity, context, slang, and many different languages, which makes it hard for machines.

- Ambiguity means one word can have many meanings, like "bank."
- Sarcasm, idioms, and humour are hard to understand.
- Low-resource languages have little training data.
- Models can inherit bias and can be tricked by prompts.
- **Real-world example:** "I saw her duck" can mean two different things.
- **Real-world example:** Indian languages with mixed English like Hinglish are hard for many models.

Q72. What is the difference between a rule-based chatbot and an AI (LLM) chatbot?

Answer: A rule-based chatbot follows fixed scripts and keywords, while an AI chatbot understands the meaning of your message and creates its own reply.

- Rule-based bots only work for questions the developer planned in advance.
- AI chatbots handle new wording, spelling mistakes, and follow-up questions.
- Rule-based bots are predictable and safe, but AI bots can make mistakes or be tricked.
- Many companies combine both: rules for sensitive steps and AI for natural conversation.
- **Real-world example:** A simple bank menu bot that says "Press 1 for balance" is rule-based.
- **Real-world example:** ChatGPT-style support assistants answer open questions, but need guardrails against prompt injection.

SECTION 5: DATA SCIENCE (DS)

Q73. What is Data Science and what is its lifecycle?

Answer: Data Science is the field of collecting, cleaning, analysing, and understanding data to find useful insights and help in decisions.

- Step 1 is to understand the business problem.
- Step 2 and 3 are collecting and cleaning the data.
- Step 4 and 5 are exploring the data and building models.
- Step 6 and 7 are evaluating, deploying, and monitoring the model.
- **Real-world example:** Swiggy and Zomato predict delivery times using data science.
- **Real-world example:** A security team analyses log data to find patterns of repeated attacks.

Q74. How do you handle missing values in a dataset?

Answer: We handle missing values by removing them, filling them with a sensible value, or using a model to predict them, depending on how much is missing and why.

- Drop rows or columns if only a few values are missing or the column is useless.
- Fill numbers with mean or median, and categories with the most common value.
- Advanced methods use KNN or model-based imputation.
- Always find out why the data is missing, because it may carry meaning.
- **Real-world example:** A hospital fills missing age values with the median age of patients.
- **Real-world example:** Missing fields in a log may themselves be a sign of a tampered record.

Q75. What are outliers and how do you handle them?

Answer: Outliers are data points that are very different from the rest, and they may be errors or important rare events.

- Find them using box plots, Z-score, or the IQR method.
- Remove them if they are clear data errors.
- Keep them when they are real, such as fraud cases.
- Alternatively, cap values or use robust methods like the median.
- **Real-world example:** A salary of 10 crore in a dataset of ordinary employees may be a data entry error.
- **Real-world example:** In security, a huge data transfer at 3 AM is an outlier that may show data theft.

Q76. What is Exploratory Data Analysis (EDA)?

Answer: EDA is the step where we study the data using summaries and charts to understand its patterns, problems, and relationships before modelling.

- We check size, data types, missing values, and distributions.
- Charts like histograms, box plots, and heatmaps are used.
- It helps to find errors, outliers, and useful features.
- It guides the choice of model.
- **Real-world example:** A retail analyst plots sales by month and finds a festival season spike.
- **Real-world example:** Analysts draw a heatmap of login attempts by hour to spot brute-force attacks.

Q77. What is the difference between normalization and standardization?

Answer: Normalization rescales data to a fixed range like 0 to 1, while standardization rescales data to have a mean of 0 and a standard deviation of 1.

- Both put features on the same scale so that no feature dominates.
- Normalization is sensitive to outliers.
- Standardization works better for many algorithms like SVM, logistic regression, and neural networks.
- Tree-based models usually do not need scaling.
- **Real-world example:** Age (0 to 100) and salary (0 to 1,00,000) must be scaled before using KNN.
- **Real-world example:** Image pixel values 0 to 255 are normalised to 0 to 1 before training a CNN.

Q78. What is the difference between correlation and causation?

Answer: Correlation means two things move together, while causation means one thing actually makes the other happen.

- A strong correlation does not prove one causes the other.
- A hidden third factor (confounder) may cause both.
- Proving causation needs controlled experiments such as A/B tests.
- Mixing them up leads to wrong business decisions.
- **Real-world example:** Ice cream sales and drowning both rise in summer, but ice cream does not cause drowning.
- **Real-world example:** Many alerts on a day does not always mean more attacks, it may just be a noisy new rule.

Q79. What is hypothesis testing and a p-value?

Answer: Hypothesis testing is a method to check if a result is real or just by chance, and the p-value tells how likely the result is if there is actually no effect.

- The null hypothesis says there is no effect, and the alternative says there is an effect.
- A p-value below 0.05 is usually treated as statistically significant.
- A small p-value does not tell the size or importance of the effect.
- Errors can be Type I (false alarm) or Type II (missed effect).
- **Real-world example:** A/B testing checks if a new website button truly increases clicks.
- **Real-world example:** A company tests if a new security awareness programme really reduced phishing clicks.

Q80. What are mean, median, mode, variance, and standard deviation?

Answer: Mean, median, and mode describe the centre of the data, while variance and standard deviation describe how spread out the data is.

- Mean is the average, median is the middle value, and mode is the most frequent value.
- Median is better than mean when there are outliers.
- Variance is the average squared distance from the mean.
- Standard deviation is the square root of variance, in the same unit as data.
- **Real-world example:** Median salary is more honest than mean salary if a few CEOs earn very high pay.
- **Real-world example:** A sudden rise in standard deviation of network traffic can show unusual activity.

Q81. What are structured, semi-structured, and unstructured data?

Answer: Structured data fits in tables with rows and columns, semi-structured data has some tags or structure, and unstructured data has no fixed format.

- Structured examples are SQL tables and Excel sheets.
- Semi-structured examples are JSON, XML, and log files.
- Unstructured examples are emails, images, audio, and videos.
- Most of the data in the world today is unstructured.
- **Real-world example:** A bank's customer table is structured data.
- **Real-world example:** Server logs in JSON are semi-structured, and CCTV footage is unstructured.

Q82. What is Big Data and what are the 5 Vs?

Answer: Big Data means data so large, fast, or varied that normal tools cannot handle it, and it is described by Volume, Velocity, Variety, Veracity, and Value.

- Volume is the huge size, and Velocity is the speed at which it arrives.
- Variety is the different formats, and Veracity is how accurate the data is.
- Value is the useful insight we get from it.
- Tools include Hadoop, Spark, and Kafka.
- **Real-world example:** Facebook and YouTube produce enormous data every second.
- **Real-world example:** A large SOC collects billions of log events per day, which is a Big Data problem.

Q83. Which Python libraries are most used in data science?

Answer: The most common libraries are NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, and PyTorch.

- NumPy handles fast numerical arrays.
- Pandas is used to clean and analyse tables of data.
- Matplotlib and Seaborn are used for charts.
- Scikit-learn offers ready ML algorithms, and TensorFlow and PyTorch are for deep learning.
- **Real-world example:** An analyst loads a CSV file in Pandas to clean sales data.
- **Real-world example:** A security analyst uses Pandas and Scikit-learn to detect anomalies in firewall logs.

Q84. What is a data pipeline and ETL?

Answer: A data pipeline moves data from sources to a destination, and ETL stands for Extract, Transform, Load, which are the three steps of that movement.

- Extract takes data from databases, files, or APIs.
- Transform cleans and reshapes the data.
- Load stores the data in a data warehouse or lake.
- Good pipelines are automated, monitored, and secure.
- **Real-world example:** A company moves daily sales data from many shops into one warehouse for reports.
- **Real-world example:** A SIEM collects logs from many systems, normalises them, and stores them for analysis.

Q85. How do data privacy and anonymisation work in data science?

Answer: Data privacy means protecting personal data, and anonymisation removes or masks details so that a person cannot be identified.

- Techniques include masking, hashing, tokenisation, and generalisation.
- Differential privacy adds small noise so that individuals are hidden.
- Laws like GDPR and India's DPDP Act require consent and protection of personal data.
- Poorly anonymised data can be linked back to a person.
- **Real-world example:** Netflix released an anonymised dataset in 2006 and researchers re-identified users by matching it with IMDb reviews.
- **Real-world example:** Hospitals remove names and phone numbers before sharing data for research.

SECTION 6: CHATGPT AND LARGE LANGUAGE MODELS

Q86. What is ChatGPT and how does it work?

Answer: ChatGPT is a chatbot by OpenAI built on GPT models that predicts the next word again and again to produce human-like answers.

- It is based on the Transformer architecture and trained on huge amounts of text.
- First it is pretrained to predict the next token, and then it is tuned to follow instructions.
- RLHF (human feedback) makes its answers more helpful and safe.
- It does not "know" facts like a database, so it may make errors.
- **Real-world example:** Students and developers use it for explaining topics and writing code.
- **Real-world example:** Companies use ChatGPT-based tools for customer support and report drafting.

Q87. What does GPT stand for?

Answer: GPT stands for Generative Pre-trained Transformer.

- Generative means it creates new text.
- Pre-trained means it first learns from a large amount of text before special tuning.
- Transformer is the neural network design it is built on.
- Each new GPT version is bigger and better at reasoning.
- **Real-world example:** GPT models power ChatGPT and many apps that use the OpenAI API.
- **Real-world example:** Microsoft Copilot uses OpenAI models inside Word, Excel, and Teams.

Q88. What is RLHF (Reinforcement Learning from Human Feedback)?

Answer: RLHF is a training step where humans rank the model's answers, and the model learns to give answers that people prefer.

- Humans compare several answers and pick the better one.
- A reward model learns these human preferences.
- The LLM is then tuned to get a high reward score.
- It improves helpfulness and safety but does not remove all errors.
- **Real-world example:** It is why ChatGPT politely refuses harmful requests.
- **Real-world example:** Without RLHF, the base model would just continue text and not answer like an assistant.

Q89. What is prompt engineering?

Answer: Prompt engineering is the skill of writing clear instructions so an AI model gives the best possible answer.

- Give a clear role, task, context, and output format.
- Few-shot prompting adds examples in the prompt.
- Chain-of-thought asks the model to reason step by step.
- Iterating and testing prompts improves results.
- **Real-world example:** "You are a SOC analyst, summarise this log in 5 bullet points" gives a better result than "summarise this."
- **Real-world example:** Companies keep a library of tested prompts for support and reports.

Q90. What are prompt injection and jailbreaking?

Answer: Prompt injection is an attack where hidden or malicious text tricks an AI into ignoring its rules, and jailbreaking is persuading the model to break its safety limits.

- Direct injection is typed by the user, and indirect injection is hidden in a web page, email, or file the AI reads.
- It is ranked as a top risk in the OWASP Top 10 for LLM Applications.
- Defences are input and output filters, least privilege, separating instructions from data, and human approval for risky actions.
- No single fix exists, so layered defence is needed.
- **Real-world example:** A web page with hidden white text told an AI browsing tool to leak the user's data.
- **Real-world example:** Early users made a chatbot reveal its secret system prompt by saying "ignore previous instructions."

Q91. What is fine-tuning of an LLM?

Answer: Fine-tuning means training an existing LLM a little more on your own data so that it behaves better for a specific task or style.

- It changes the model's weights, unlike prompting or RAG.
- Good for fixed style, tone, or special formats.
- It needs quality data and costs time and money.
- It is not the best way to add fresh facts, because RAG is better for that.
- **Real-world example:** A law firm fine-tunes a model to write in its legal drafting style.
- **Real-world example:** A company trains a model on its own ticket data to classify support issues.

Q92. What are the data privacy risks of using ChatGPT at work?

Answer: Anything typed into a public AI tool may leave your company's control, so employees can accidentally leak confidential data.

- Data may be stored and used to improve models, depending on the plan and settings.
- Enterprise plans and private deployments offer stronger data controls.
- Companies should classify data and block sensitive data from being pasted.
- Staff need AI usage training and clear policy.
- **Real-world example:** Samsung engineers pasted internal source code into ChatGPT in 2023, and the company then restricted its use.
- **Real-world example:** Many banks and firms blocked public chatbots until they had approved enterprise versions.

Q93. What do temperature and top-p mean in LLMs?

Answer: Temperature and top-p are settings that control how random or creative the model's answer is.

- Low temperature (near 0) gives safe and consistent answers.
- High temperature gives more varied and creative answers.
- Top-p picks from the smallest group of likely words whose probabilities add up to p.
- Use low values for facts and code, and higher values for creative writing.
- **Real-world example:** A customer support bot uses low temperature for consistent answers.
- **Real-world example:** A poem or ad copy generator uses a higher temperature for creativity.

SECTION 7: CLAUDE (ANTHROPIC)

Q94. What is Claude?

Answer: Claude is a family of AI assistants made by Anthropic, a company focused on building safe and helpful AI.

- It is a large language model that can chat, write, analyse documents, and code.
- It is available through the Claude app, the web, and an API for developers.
- Anthropic was founded in 2021 by former OpenAI researchers.
- It is designed to be helpful, honest, and harmless.
- **Real-world example:** People use Claude to summarise long reports and contracts.
- **Real-world example:** Developers use Claude to write and review code.

Q95. What is Constitutional AI?

Answer: Constitutional AI is Anthropic's training method where the model is guided by a written set of principles (a constitution) to improve its own answers and be safer.

- The model critiques and revises its answers using these principles.
- It reduces the need for humans to label many harmful outputs.
- Principles cover things like avoiding harm and being honest.
- It makes the model's values more clear and open to review.
- **Real-world example:** Claude can politely refuse a request to build malware while still explaining the safe alternative.
- **Real-world example:** It is one reason Claude is often chosen by firms that care about safety and compliance.

Q96. What are the Claude model families (Opus, Sonnet, Haiku)?

Answer: Anthropic offers Claude in different sizes, where Opus is the most capable, Sonnet is the balanced one, and Haiku is the fastest and cheapest.

- Bigger models are better at hard reasoning but cost more and respond slower.
- Smaller models are great for simple, high-volume tasks.
- You pick the model based on speed, cost, and quality needed.
- New versions are released often, so check the latest model list.
- **Real-world example:** A company uses Haiku to quickly sort thousands of support emails.
- **Real-world example:** A research team uses Opus or Sonnet for deep analysis and complex coding.

Q97. What is the difference between Claude and ChatGPT?

Answer: Both are LLM chat assistants, but they are made by different companies, Anthropic and OpenAI, with different training approaches, features, and strengths.

- Claude uses Constitutional AI, and ChatGPT uses RLHF heavily.
- Claude is well known for handling long documents and careful writing.
- ChatGPT has a wide ecosystem with plugins, image generation, and custom GPTs.
- Strengths change with every new release, so test both on your own task.
- **Real-world example:** Many developers use Claude Code for working in a codebase.
- **Real-world example:** Many businesses use ChatGPT through Microsoft Copilot.

Q98. What is Claude Code?

Answer: Claude Code is an AI coding assistant from Anthropic that works in your terminal or code editor and can read your project, edit files, and run commands.

- It is an example of agentic AI because it plans and takes actions.
- It can fix bugs, write tests, and explain unfamiliar code.
- Developers can approve or reject each action for safety.
- Using it safely means giving it limited permissions and reviewing changes.
- **Real-world example:** A developer asks it to find and fix a failing test in a large project.
- **Real-world example:** A security engineer asks it to review code for common vulnerabilities, then checks its findings.

Q99. What is MCP (Model Context Protocol)?

Answer: MCP is an open standard introduced by Anthropic that lets AI assistants connect to tools and data sources, such as files, databases, and apps, in one common way.

- Before MCP, each tool needed its own custom connection.
- An MCP server exposes tools, and an AI app (client) uses them.
- It is like a USB-C port for connecting AI to many systems.
- It brings security risks, so use trusted servers and limited permissions.
- **Real-world example:** An AI assistant uses an MCP server to read tickets from Jira or files from Google Drive.
- **Real-world example:** Malicious or poorly secured MCP servers can enable data leakage or prompt injection, which is a growing security topic.

Q100. How can Claude or other LLMs help in cybersecurity, and what are the limits?

Answer: LLMs can speed up security work like summarising alerts, explaining code, and writing reports, but humans must verify the results because LLMs can be wrong or tricked.

- Uses: alert triage, log explanation, writing detection rules, policy drafts, and security awareness content.
- They help junior analysts learn faster.
- Limits are hallucination, prompt injection, and data privacy concerns.
- Never paste sensitive data into unapproved tools, and keep a human in the loop.
- **Real-world example:** A SOC analyst asks an LLM to explain a suspicious PowerShell command and gets a quick breakdown.
- **Real-world example:** Attackers also misuse LLMs for phishing text, so defenders need AI-aware detection too.

SECTION 8: RAG, LANGCHAIN AND LANGGRAPH (QUICK ROUND)

Q101. What is RAG (Retrieval-Augmented Generation)?

Answer: RAG is a method where the AI first searches a trusted knowledge source for relevant information and then uses it to write a more accurate answer.

- It adds fresh and private company data to an LLM without retraining it.
- It reduces hallucination because answers are based on real documents.
- It can show the source of the answer, which helps trust and audit.
- The main parts are a retriever and a generator (LLM).
- **Real-world example:** A company chatbot answers HR policy questions from its own policy PDFs.
- **Real-world example:** A SOC assistant answers questions using the company's runbooks and past incident reports.

Q102. How does a RAG system work step by step?

Answer: RAG splits documents into chunks, converts them to embeddings, stores them, finds the best matching chunks for a question, and gives them to the LLM to answer.

- Step 1: Load and split documents into small chunks.
- Step 2: Convert each chunk into an embedding vector.
- Step 3: Store the vectors in a vector database.
- Step 4: Convert the user's question into a vector and find the closest chunks.
- Step 5: Send the question and the chunks to the LLM, which writes the final answer.
- **Real-world example:** Asking "What is our leave policy?" retrieves the leave policy paragraph and the LLM explains it simply.

Q103. What are embeddings and a vector database?

Answer: Embeddings are number lists that capture the meaning of text, and a vector database stores them so we can quickly find texts with similar meaning.

- Similar meanings give vectors that are close to each other.
- Search uses similarity measures like cosine similarity.
- Popular vector databases are Pinecone, FAISS, Chroma, and Weaviate.
- Access control on the database is important, because it holds company data.
- **Real-world example:** Searching "reset my password" can also find a document titled "account recovery steps."
- **Real-world example:** Enterprise search tools use vector databases to search across thousands of internal documents.

Q104. What is the difference between RAG and fine-tuning?

Answer: RAG gives the model outside knowledge at question time, while fine-tuning changes the model itself through extra training.

- RAG is better for facts that change often, because you just update the documents.
- Fine-tuning is better for style, tone, or special output formats.
- RAG is cheaper and easier to update and also gives sources.
- Many real systems use both together.
- **Real-world example:** A news chatbot uses RAG so it can answer using today's articles.
- **Real-world example:** A brand voice assistant uses fine-tuning to always write in the company's style.

Q105. What is LangChain?

Answer: LangChain is an open-source framework that helps developers build applications powered by LLMs by connecting models, prompts, data, and tools.

- It offers ready building blocks, so we do not write everything from scratch.
- It works with many LLMs such as OpenAI, Claude, and open-source models.
- It makes RAG chatbots and AI agents easier to build.
- It is available for Python and JavaScript.
- **Real-world example:** A developer builds a "chat with your PDF" app using LangChain.
- **Real-world example:** A team builds a customer support bot that searches documents and creates tickets.

Q106. What are the key components of LangChain?

Answer: The main components are models, prompt templates, chains, memory, retrievers, tools, and agents.

- Models are the LLMs or chat models.
- Prompt templates are reusable prompts with blanks to fill.
- Chains join steps together, and memory remembers the earlier chat.
- Retrievers fetch documents, and tools let the model call APIs or run code.
- Agents let the LLM decide which tool to use and when.
- **Real-world example:** A travel bot uses a prompt template, a flight search tool, and memory to remember your trip dates.

Q107. What is LangGraph?

Answer: LangGraph is a framework, built by the LangChain team, to create AI agents and workflows as a graph of steps with loops, branches, and saved state.

- Each node is a step, such as calling an LLM or a tool, and edges decide what happens next.
- It supports loops, so an agent can retry or reflect until the task is done.
- It keeps state, so long tasks can pause and resume.
- It supports human-in-the-loop approvals before risky actions.
- **Real-world example:** A support agent that checks the order, decides to refund or escalate, and asks a human to approve large refunds.
- **Real-world example:** A security triage agent that gathers logs, scores risk, and asks an analyst before isolating a machine.

Q108. What is the difference between LangChain and LangGraph?

Answer: LangChain is good for simple, mostly straight-line workflows, while LangGraph is better for complex agents that need loops, branching, and control.

- LangChain chains run in a fixed order from start to end.
- LangGraph lets you design the exact flow, including cycles and conditions.
- LangGraph gives better control, debugging, and state handling for production agents.
- They are often used together, since LangGraph builds on the LangChain ecosystem.
- **Real-world example:** A simple "summarise this document" app is easy with LangChain.
- **Real-world example:** A multi-agent research assistant that plans, searches, checks, and rewrites is better in LangGraph.

Q109. What are chunking and the common challenges in building a RAG system?

Answer: Chunking means cutting documents into small parts for search, and the challenge is that bad chunking or poor retrieval leads to wrong or incomplete answers.

- Chunks that are too big add noise, and chunks that are too small lose context.
- Overlapping chunks help keep the meaning between parts.
- The retriever may fetch irrelevant text, so re-ranking and hybrid search (keyword plus vector) help.
- Outdated or messy documents give bad answers, so data quality matters.
- **Real-world example:** A bot split a contract in the middle of a clause and gave an incomplete answer.
- **Real-world example:** Teams test RAG with sample questions to measure how often the right chunk is found.

Q110. What are the security risks in RAG and AI agent applications?

Answer: The main risks are indirect prompt injection, data leakage, excessive permissions, and poisoned documents.

- A malicious document in the knowledge base can carry hidden instructions to the LLM.
- Users may see documents they should not, if access control is missing.
- Agents with too many permissions can be tricked into harmful actions like sending data out.
- Defences are access control on retrieval, input and output filters, least privilege, logging, and human approval.
- **Real-world example:** A hidden instruction in a shared document told an AI assistant to email private data to an attacker.
- **Real-world example:** Following the OWASP Top 10 for LLM Applications is a good starting checklist for such systems.
